

Model review and revisions

JMP progress, September 17–October 2, 2026

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I reviewed the full model and calibration: preferences, borrowing rules, demographic accounting, equilibrium conditions, empirical measurements, targets, and the estimation procedure.

- Returned to a Stone–Geary housing requirement that applies only to parents, retaining nonlinear benefits from children.
- Reviewed equivalence scales and housing-share alternatives.
- Rebuilt the first-birth housing evidence and national input measures.
- Checked demographic accounting, credit, and adult entry; recalibrated the model.

Household decisions

Overlapping generations of households differ in age, earnings, wealth, housing tenure, and family size.

- During fertile ages, households choose whether to attempt a birth; success becomes less likely with age.
- They choose consumption, saving, housing size, and whether to rent or own; renters pay rent, while owners make a down payment and face selling costs.
- Persistent earnings risk and borrowing limits shape saving; a first-birth fixed cost and taste shocks shape fertility.
- Households retire, face mortality risk, and value bequests.

The interaction of housing costs and available wealth affects both birth timing and tenure choice.

Children and housing needs

The first child requires basic space at home; each additional child raises the goods needs of a larger family.

$$u(c, s; m) = \frac{\left[\frac{c^{\alpha_0} s^{1-\alpha_0}}{e(m)} \right]^{1-\sigma}}{1-\sigma} + \psi m^{1-\gamma},$$
$$e(m) = \left(\frac{2 + 0.7m}{2} \right)^{0.7}, \quad s = \eta_\tau [h - h_P \mathbf{1}\{m > 0\}],$$
$$\eta_R = 1, \quad \eta_O = \chi.$$

Here h is physical rooms, m is children at home, and $\tau \in \{R, O\}$ denotes renting or owning. Parents choose $h > h_P$, and α_0 is the constant consumption exponent. The equivalence scale raises goods needs with family size; the benefit is zero at $m = 0$, with ψ setting its level and γ its curvature so the marginal benefit diminishes as m grows.

Demographic and market equilibrium

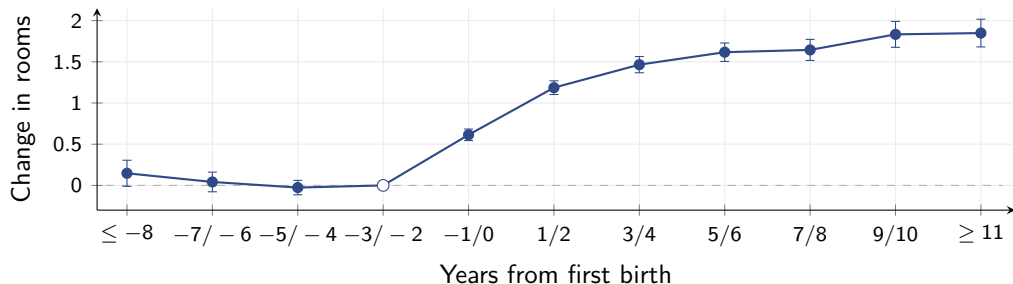
Children leaving home and new adult households entering the economy are separate transitions.

- Children leave dependency independently; adult entry follows earlier births with a lag.
- Positive estates finance entrant assets; any remainder leaves the economy.
- Stationary calibration normalizes population to one: the house price satisfies birth renewal, and housing clearing determines H_0 .
- A balanced payroll tax finances equal retiree pensions.

A stationary equilibrium has constant prices, allocations, and cohort structure, with demographic reproduction and market clearing.

Housing around the first birth

I rebuilt the PSID event study using interview dates and household status before the first birth.



At years 3–4, rooms rise by **1.465** (SE 0.050), ownership by **26.886 pp**, and recent moving falls by **21.147 pp**.

Reference person or spouse before birth. Baseline: years $-3/-2$; person and year effects; survey weights; 95% pointwise intervals.

Calibration strategy and inputs

I estimate a 2007 stationary economy using national measures of housing, earnings, wealth, and fertility.

Input	Value	Measurement
Mean occupied rooms	5.729	AHS 2007
Pension / working-household earnings	0.229	CPS ASEC, income 2007
Annual child-directed estates / wealth	0.729%	SCF 2007 mortality proxy
Annual depreciation / property tax	1.416% / 1.060%	National housing inputs

- Ten parameters are estimated jointly, including ψ . Other exogenous inputs include earnings and demographics, housing depreciation and taxes, the interest rate, and mortgage financing.
- The objective is a weighted sum of squared gaps for the original ten scored moments, using working error scales.
- Three further moments are validation checks. Completed fertility is reported separately as a renewal check.

For each parameter vector, I solve household choices and birth renewal, then derive H_0 so housing demand clears at normalized population $N_0 = 1$.

The first-birth housing comparison uses a model proxy for the empirical event study.

Identification

Ten parameters are estimated jointly from ten scored moments.

Parameter	Economic interpretation	Main moments
β	Patience	Wealth / earnings
θ_0	Bequest weight	Annual bequests / wealth
χ	Owner housing-service premium	Ownership at ages 30–55; mean rooms
κ_T	Tenure tastes	Ownership at ages 30–55; recent-parent ownership gap
h_P	Basic space for parents	First-birth rooms; mean rooms
ξ	First-birth cost	Childlessness at ages 40–44; children at age 25
κ_1	First-birth taste dispersion	Mean first-birth age; children at age 25
κ_C	Later-birth taste dispersion	One-child share among mothers; children at age 25
ψ	Child-benefit level	Childlessness; children at age 25
γ	Child-benefit curvature	One-child share among mothers; children at age 25

These are joint sources of variation, not one-to-one restrictions. H_0 is derived from housing clearing; θ_1 is fixed externally. Validation moments appear separately; local identification rank is not yet verified.

Stone–Geary calibration

2007 stationary economy; 80% mortgage financing

Moment	Target	Soft*	Hard	Quarter-saving
Completed fertility (renewal check)	2.100	2.100	2.100	2.100
Childlessness, ages 40–44 (%)	19.828	20.222	20.413	20.546
One child among mothers, ages 40–44 (%)	21.366	21.264	20.907	20.844
Mean first-birth age (years)	25.976	26.032	26.007	26.057
Wealth / earnings	6.927	6.551	6.610	6.645
Annual child bequests / wealth (%)	0.729	0.685	0.684	0.685
Mean occupied rooms	5.729	5.957	6.241	6.131
Ownership, ages 30–55 (%)	67.626	66.527	60.878	64.572
First-birth rooms difference	1.465	1.324	1.063	1.167
Recent-parent ownership difference (pp)	12.761	11.825	10.125	11.683
Children at age 25 (capped at three)	0.810	0.528	0.524	0.521
Validation moments	Target	Soft*	Hard	Quarter-saving
First births at age 30+ (%)	24.928	23.448	23.078	23.395
Older wealth/income p90/p50	3.516	2.968	2.861	2.937
Family rooms difference	0.385	0.304	0.321	0.316

Soft*: best saved search point. Hard and quarter-saving: final numerical postchecks passed.

Household policies and lifecycle paths

Housing supply and borrowing constraints

Transition dynamics

Parameter estimates

Ten parameters are estimated jointly; H_0 is derived from housing-market clearing.

Parameter	Economic interpretation	Soft*	Hard	Quarter-saving
β	Annual discount factor	0.967	0.967	0.968
χ	Owner housing-service premium	1.097	1.068	1.091
ξ	First-birth fixed utility cost	0.352	0.385	0.397
κ_1	First-birth taste dispersion	0.124	0.123	0.128
κ_C	Later-birth taste dispersion	0.363	0.350	0.348
θ_0	Bequest weight	0.106	0.106	0.109
γ	Child-benefit curvature	0.101	0.101	0.098
κ_T	Tenure-choice dispersion	0.014	0.014	0.015
ψ	Child-benefit level	0.178	0.172	0.174
h_P	Basic space requirement for parents (rooms)	2.504	2.562	2.492
H_0	Housing supply coefficient (derived)	6.757	7.546	7.289

H_0 is derived at normalized population $N_0 = 1$. Fixed inputs: $\sigma = 2$; $\alpha_0 = 0.733$; annual interest 2%; financed share $\phi = 0.8$. Full parameter bounds are in the saved tables.

Credit timing

With $\phi = 0.8$, a purchase requires a 20% down payment. Purchase timing determines which resources count toward it. The soft rule includes current-period income in the purchase screen; the other rules use:

$$\text{Hard: } A \geq (1 - \phi)Q,$$

$$\text{Quarter-saving: } A + \frac{1}{4} [b' - (A - Q)] \geq (1 - \phi)Q.$$

Here A is financial wealth plus net proceeds from selling the previous home, Q is the cost of the chosen home, and b' is the end-of-period financial balance. Net saving is $b' - (A - Q)$.

For a home costing 100 with 10 available at closing, the hard rule blocks the purchase. The quarter-saving rule meets the 20-unit equity requirement if net saving is 40:

$$10 + 0.25 \times 40 = 20.$$

This approximates saving before purchase within a four-year period. Under quarter-saving, income helps only if saved. Hard and quarter-saving also impose an end-of-period net-estate floor for buyers; this is an additional change from soft.

Calibration comparison

- The soft rule gives the best saved fit. All three calibrations use the same targets and weights.
- Mean rooms: 5.957 (soft), 6.241 (hard), 6.131 (quarter-saving), against a target of 5.729.
- First-birth rooms difference: 1.324, 1.063 and 1.167, respectively, against 1.465.
- Children at age 25: 0.528, 0.524 and 0.521, against 0.810. Early fertility remains low in all three.
- Each specification is recalibrated. The comparison combines purchase timing, the buyer net-estate restriction and parameter adjustment.